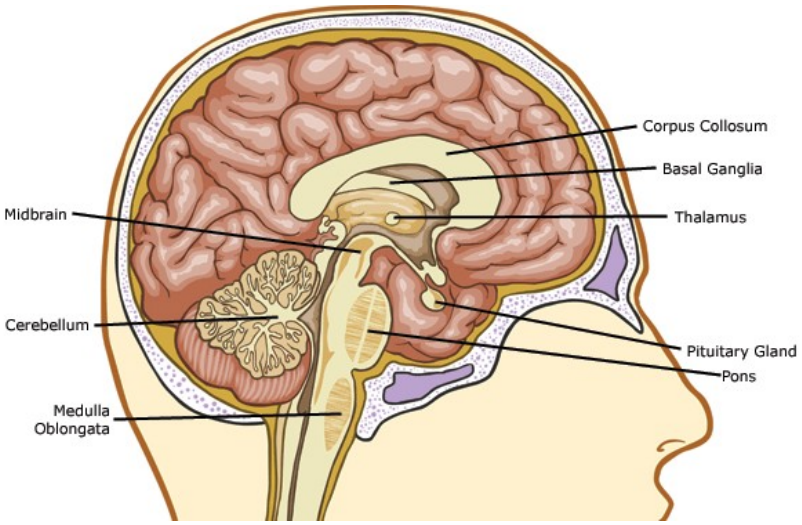
Transsphenoidal Hypophysectomy *Prodigy Anesthesia*

A transsphenoidal hypophysectomy is performed to remove tumors of the pituitary gland. The operation is carried out through the gingival mucosa below the upper lip, across the nasal septum, and through the superior aspect of the sphenoid sinus into the sella turcica, where the pituitary gland is located.

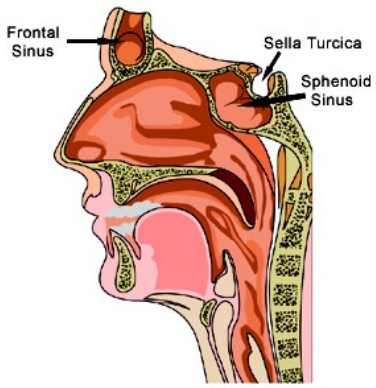
Pituitary adenomas are the most common neoplasms treated with this procedure, followed by craniopharyngiomas and parasellar meningiomas. Malignant and/or metastatic tumors of the pituitary gland are rare.

Functional tumors (tumors that secrete hormones) are typically detected early because of their systemic effects whereas other tumors are



detected later as symptoms of increased intracranial pressure or compression of nearby structures occurs.

The anterior pituitary secretes adrenocorticotropic hormone (ACTH), thyroid stimulating hormone, growth hormone, follicle-stimulating hormone, luteinizing hormone, and prolactin. The posterior pituitary releases antidiuretic hormone and oxytocin. The most common type of functional pituitary tumor produces hyperprolactinemia. The release of prolactin by the tumor can cause amenorrhea, galactorrhea, or both, in women. In mean, it can result in galactorrhea or infertility.



Adenomas that secrete ACTH can produce symptoms of Cushing's disease such as truncal obesity, moon-face, proximal muscle weakness, hypertension, and impaired glucose tolerance. Excessive release of growth hormone can result in gigantism, osteoarthritis, diabetes, hypertension, and heart disease. Patients presenting with symptoms of an expanding mass in the brain can present with increase intracranial pressure (headache, nausea, and vomiting) or visual disturbances as the mass encroaches upon the optic chiasm. Diabetes insipidus can occasionally be seen preoperatively if the tumor involves the posterior pituitary.

The traditional anesthetic techniques for craniotomy are employed for a transsphenoidal

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hypophysectomy. Antibiotic coverage is necessary and administration of a glucocorticoid is routine in most facilities. Most anesthetists avoid nitrous oxide to prevent any problems stemming from a pneumocephalus. Large-bore IVs are needed and a central line and precordial doppler may be considered for detection and treatment of a venous air embolism as the head is often elevated for this procedure.

Complications related to this procedure may arise from the need for mucosal injection of epinephrine to prevent bleeding, the accumulation of blood in the pharynx and stomach, risk of injury to the cavernous sinus or internal carotid artery, damage to cranial nerves III and IV and the branches of the trigeminal nerve that travel near the pituitary gland, and hypofunction of the pituitary gland. Diabetes insipidus develops in up to 31% of all patients but it typically only temporary.

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